wcEcoli and the 4D Whole-Cell Model: Background and Comparison

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March 2026

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1 What is wcEcoli and Why Am I Using It?

1.1 What is wcEcoli?

wcEcoli is an open-source whole-cell computational model of *Escherichia coli*, developed by the Covert Lab at Stanford University. It is the most complete whole-cell model of a free-living organism currently available.

The project descends from the first-ever whole-cell model, published in Karr et al. (2012, *Cell*), which simulated the entire life cycle of *Mycoplasma genitalium* — a minimal bacterium with only ~470 genes. The *E. coli* model is substantially more complex: it covers roughly 43% of the organism's characterised genes and integrates 17 distinct biological processes running simultaneously.

Key references:

- Karr JR et al. (2012) *A whole-cell computational model predicts phenotype from genotype.* Cell 150(2):389–401. https://doi.org/10.1016/j.cell.2012.05.044
- Sun G, Ahn-Horst TA, Covert MW (2021) *The E. coli whole-cell modeling project.* EcoSal Plus 9(2). https://doi.org/10.1128/ecosalplus.ESP-0001-2020
- Ahn-Horst TA et al. (2022) *An expanded whole-cell model of E. coli links cellular physiology with mechanisms of growth rate control.* npj Syst Biol Appl 8:30. https://doi.org/10.1038/s41540-022-00242-9

1.2 What does it simulate?

The model simulates a single *E. coli* cell from birth to division, advancing in one-second timesteps. It tracks every major cellular process simultaneously:

- **Metabolism** — flux-balance analysis over the full metabolic network; glucose uptake, amino acid synthesis, energy production
- **Transcription** — RNA polymerase binding, transcript initiation and elongation, mRNA degradation
- **Translation** — ribosome assembly, polypeptide elongation, protein degradation
- **DNA replication** — replisome positioning, origin firing, chromosome segregation
- **Gene regulation** — transcription factor binding/unbinding, two-component signalling, ppGpp-mediated stringent response
- **Cell division** — mass-based division trigger, daughter cell initialisation

Each process uses the mathematical formalism best suited to it (stochastic for gene regulation, ODE for tRNA charging, FBA for metabolism), and all share a common molecular inventory that is partitioned and reconciled at every timestep.

Output is one-second-resolution time series for every tracked quantity: ~18 listener modules recording mass fractions, metabolic fluxes, ribosome counts, replication fork positions, amino acid levels, and more.

1.3 Why is it scientifically significant?

Whole-cell models represent a qualitative shift from conventional pathway modelling. Rather than asking "what does this pathway do in isolation?", a whole-cell model asks "what does the cell do as a system?" This matters because:

- **Emergent behaviour.** The model reproduces the bacterial growth law (ribosome fraction scales linearly with growth rate) as an emergent consequence of its biochemistry, not as a built-in parameter. This kind of validation — predicting something that was not explicitly encoded — is the hallmark of a mechanistically complete model.
- **Genotype-to-phenotype.** Gene knockouts, nutrient shifts, and antibiotic perturbations can be simulated and their system-wide consequences observed, including unexpected off-target effects.
- **Integration across scales.** A single simulation spans molecular events (individual ribosome translation steps) and cell-level outcomes (doubling time, growth rate) without requiring separate models at each scale.

1.4 Why am I using it?

I am using wcEcoli as the starting point for a proposed whole-cell model of the human platelet. The rationale has two parts: scientific and practical.

1.4.1 Scientific rationale

Platelets are clinically important but computationally understudied. Existing computational models of platelet biology focus on individual pathways — calcium signalling, integrin activation, coagulation — but no model integrates them into a single, self-consistent simulation. A whole-cell model would allow questions such as:

- How does the kinetics of receptor activation propagate through calcium signalling to granule release?
- How does platelet energetics (ATP supply) constrain the speed or completeness of activation?
- What explains heterogeneity in activation thresholds across individual platelets?

These are precisely the kinds of cross-pathway, multi-scale questions that whole-cell modelling is designed to address.

1.4.2 Practical rationale

Building a whole-cell modelling framework from scratch is a multi-year engineering effort. wcEcoli provides a mature, well-tested foundation:

- The codebase cleanly separates a **generic simulation engine** (`wholecell/`) from the **organism-specific model** (`models/ecoli/`). The engine — simulation loop, state management, I/O, listener framework — is organism-agnostic by design.
- The engine already handles the technical challenges that make whole-cell modelling hard: mixed mathematical formalisms within one simulation, discrete molecule tracking, parallel process execution with shared state, efficient binary I/O at one-second resolution.

- The web UI I have built works with any organism namespace, so a platelet model would immediately have a browser-based interface for running and inspecting simulations.

The practical cost of using wcEcoli is learning a large and complex codebase. The benefit is avoiding years of framework engineering and starting with infrastructure that has already been validated against decades of *E. coli* experimental data.

1.5 Current status

I have the model running locally and on Azure. The web UI (on the `webapp` branch) is functional and actively developed. I have conducted a detailed reusability analysis of every wcEcoli component and documented the engine changes needed for a non-dividing cell type.

The next step is building a minimal platelet stub — a skeleton `models/platelet/` namespace that runs through the existing engine, produces output visible in the web UI, and can be incrementally extended with real biology.

2 The 4D Whole-Cell Model of JCVI-syn3A (MC4D)

Source paper: Thornburg ZR, Maytin A, Kwon J, Brier TA, Gilbert BR, Luthey-Schulten Z et al. "Bringing the genetically minimal cell to life on a computer in 4D." *Cell*, published 9 March 2026. DOI: 10.1016/j.cell.2026.02.009

Code repository: Luthey-Schulten-Lab/Minimal_Cell_4DWCM

2.1 What it is

MC4D is a whole-cell computational model of **JCVI-syn3A**, a synthetically constructed minimal bacterium with approximately 473 genes — one of the simplest organisms capable of independent replication. The model simulates a complete ~100-minute cell cycle, capturing every major biological process across three spatial dimensions and time (hence "4D").

This is the first whole-cell model to integrate full spatial resolution with a complete biological description of cell growth and division. Where earlier whole-cell models (including the original *M. genitalium* model and wcEcoli) treat the cell interior as well-mixed compartments, MC4D tracks where molecules are and how they move.

2.2 The organism: why JCVI-syn3A?

JCVI-syn3A is a stripped-down synthetic cell derived from *Mycoplasma mycoides* through iterative gene removal, retaining only the minimal set of genes needed for growth and division. With ~473 genes, it is dramatically simpler than *E. coli* (~4,300 genes). This makes it tractable for a model that must account for every gene product — there are simply fewer things to include.

The simplicity comes with a cost: roughly one third of syn3A's genes are of unknown function, which the model must accommodate with placeholder or inferred behaviour.

2.3 The four computational methods

The model is named "4D" both for its spatial dimensions and for the four distinct mathematical formalisms it integrates simultaneously.

2.3.1 1. RDME — Reaction-Diffusion Master Equation

Implemented via: Lattice Microbes (GPU-accelerated)

The cell interior is discretised onto a 3D lattice. Molecules are represented as particles that diffuse stochastically between lattice voxels and react with each other based on local concentrations. This captures spatial heterogeneity: the fact that two molecules can only react if they are in the same place at the same time.

Used for: the bulk of intracellular chemistry — protein synthesis (translation), molecular diffusion, bimolecular reactions, ribosome dynamics with excluded-volume constraints.

2.3.2 2. CME — Chemical Master Equation

Implemented as global stochastic reactions

A well-mixed stochastic formalism for reactions that are best treated globally rather than spatially. Used for tRNA charging reactions and transcription events, where the assumption of spatial homogeneity is reasonable and the stochastic discrete-molecule treatment matters (low copy numbers).

2.3.3 3. ODE — Ordinary Differential Equations

Implemented via: odecell framework

Deterministic kinetic integration of the metabolic network. Enzyme concentrations are extracted from the RDME simulation and passed into the ODE solver, so metabolic fluxes reflect the actual current protein composition of the cell. Outputs include dNTP pool concentrations, which are fed back to control DNA replication rates.

2.3.4 4. BD — Brownian Dynamics

Implemented via: btree_chromo with Kokkos-enabled LAMMPS

Explicit physical simulation of chromosomal DNA as a polymer. The chromosome is represented as a chain of beads subject to thermal fluctuations, constrained to the RDME lattice. Structural maintenance of chromosome (SMC) proteins (analogous to condensin) and topoisomerases control chromosome compaction and segregation. This is the most computationally demanding component and requires the LAMMPS molecular dynamics engine.

2.4 How the four methods communicate

Integration is managed by a central Python module, `Hook.py`. This module periodically interrupts the RDME simulation to:

1. Extract current enzyme and molecular counts from the RDME lattice
2. Pass enzyme concentrations to the ODE solver; run metabolic integration
3. Pass transcription events and tRNA status to the CME solver; advance gene expression
4. Synchronise chromosome state with the BD simulation
5. Write updated molecular counts back to the RDME lattice

This coupling means that metabolism, gene expression, and chromosome dynamics all influence each other dynamically throughout the simulation, rather than being run in isolation.

2.5 Biological processes covered

Process	Method	Key detail
Metabolism	ODE	Kinetic network; dNTP output constrains replication
Transcription	CME	Global stochastic; mRNA production and degradation
Translation	RDME	Spatially resolved; ribosome excluded-volume
tRNA charging	CME	Global stochastic
DNA replication	BD + ODE	Chromosome polymer dynamics; rate set by dNTP pools
Chromosome segregation	BD	SMC and topoisomerase proteins modelled explicitly
Membrane growth	RDME	Lipid and membrane protein insertion drives morphology
Cell division	RDME + BD	Triggered by chromosome segregation completion
Molecular diffusion	RDME	All major cytoplasmic species

2.6 Spatial resolution and scale

- **Lattice**: 3D voxel grid representing the cell interior
- **Cell morphology**: initialised as a sphere; morphological change during growth constrained by fluorescence imaging data from experiments
- **Timescale**: full ~100-minute cell cycle
- **Stochasticity**: because of the stochastic formalisms (RDME, CME), each simulation replicate produces a unique cell — population heterogeneity emerges naturally

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2.7 Validation

The model is validated against a wide array of experimental measurements:

- **Doubling time** — reproduced quantitatively
- **Origin-to-terminus ratio** — validated against DNA sequencing data
- **mRNA half-lives** — matched to experimental measurements
- **Protein distributions** — spatial and abundance patterns compared to fluorescence data
- **Ribosome counts** — quantitative agreement
- **Daughter cell heterogeneity** — stochastic partitioning of molecules matches experimentally observed variability between siblings

The authors emphasise that "assimilation of a wide array of experiments is necessary for construction and validation" — the model is tightly constrained by data at multiple levels.

2.8 Software and technical requirements

Component	Software	Notes
RDME solver	Lattice Microbes	GPU required
ODE integrator	odecell	Python-based
Chromosome dynamics	btree_chromo + LAMMPS	Kokkos-enabled build required
Chromosome initialisation	sc_chain_generation	Generates initial polymer configs
Membrane (optional)	FreeDTS	For membrane shape analysis
Orchestration	Hook.py (custom)	Central integration module

Running a full simulation requires a GPU (for Lattice Microbes) and a correctly configured LAMMPS build with Kokkos support. The model runs as a single Python entry point (`Whole_Cell_Minimal_Cell.py`) with parameters for simulation duration, output directory, GPU device selection, and random seeds for reproducibility.

Input data lives in `input_data/` and includes kinetic parameters (xlsx), genomic sequences (GenBank), metabolic network definitions (SBML), and DNA polymer properties.

2.9 Key scientific contribution

The central claim of the paper is that spatial heterogeneity within the cell — the fact that molecules are not uniformly distributed — materially affects the biochemical reactions that control cellular phenotype. By resolving this spatial structure, MC4D can predict behaviours that well-mixed models cannot, including the spatial patterns of protein synthesis, the physical dynamics of chromosome segregation, and the sources of cell-to-cell variability in division outcomes.

This is a significant step beyond the original *M. genitalium* whole-cell model (which used the same organism's precursor but had no spatial resolution) and beyond wcEcoli (which achieves greater biological coverage but treats the cell as a set of well-mixed compartments).

2.10 Limitations and context

- Requires specialist software dependencies and GPU hardware — harder to run than wcEcoli
- The organism (JCVI-syn3A) is simpler than any naturally occurring bacterium; many results may not generalise directly
- ~1/3 of syn3A genes have unknown function; the model must make assumptions about these
- No public web interface or high-level user tooling; the model is a research prototype
- Computational cost is substantially higher than non-spatial whole-cell models

3 wcEcoli vs MC4D: A Comparison of Two Whole-Cell Modelling Approaches

These are the two most complete whole-cell models currently available. They share a common ambition — simulate an entire cell — but make substantially different choices about organism, formalism, and what "complete" means. Understanding the differences matters for deciding which approach, or which combination, is most useful for extending to human platelets.

3.1 Side-by-side overview

Dimension	wcEcoli	MC4D
Organism	*E. coli* (wild-type, ~4,300 genes)	JCVI-syn3A (~473 genes, synthetic minimal)
Gene coverage	~43% of characterised genes	~100% (all ~473 genes, including unknowns)
Spatial resolution	None — well-mixed compartments	Full 3D (RDME lattice + BD chromosome)
Cell cycle	~25–130 min depending on condition	~100 min
Metabolism	Flux-balance analysis (FBA)	ODE kinetic network
Transcription	Stochastic (CME-like), well-mixed	CME, global stochastic
Translation	Stochastic, well-mixed	RDME, spatially resolved
Chromosome	Discrete replisome tracking (position only)	Brownian dynamics polymer (LAMMPS)
Formalisms	FBA + stochastic + ODE	RDME + CME + ODE + BD
Stochasticity	Partial (gene regulation, translation)	Full (all molecular processes)
GPU required	No	Yes (RDME via Lattice Microbes)
Web interface	Yes (custom Dash app)	None
Framework reuse	Explicitly designed for it	Research prototype
Published	npj Sys Bio Appl, 2022	Cell, March 2026
Lab	Covert Lab, Stanford	Luthey-Schulten Lab, UIUC

3.2 Organism choice: complexity vs completeness

This is the most fundamental difference. wcEcoli models *E. coli* — a well-characterised, non-minimal organism with thousands of genes and decades of experimental data. The trade-off is that the model can only cover ~43% of genes; the rest are either not yet modelled or not characterised well enough to include.

MC4D models JCVI-syn3A, a synthetic cell engineered to be as simple as possible. With ~473 genes, it is feasible — in principle — to include every gene product. The trade-off is that roughly a third of those genes have unknown function, and the organism is too stripped-down to exhibit many of the regulatory behaviours (stress responses, growth-rate adaptation) that make *E. coli* biologically interesting.

For platelet modelling: Neither organism is a direct analogue, but the wcEcoli philosophy — deep coverage of a complex, well-studied cell — is more relevant. Platelets are well characterised biochemically; the goal is integrating known biology, not filling in unknowns.

3.3 Spatial resolution: the central architectural divide

wcEcoli treats the cell as a set of well-mixed compartments (cytoplasm, periplasm, extracellular). Molecules exist as integer counts, not positions. This is computationally efficient and sufficient for many questions, but it cannot address anything that depends on *where* a molecule is.

MC4D places every molecule on a 3D lattice and simulates diffusion explicitly. This reveals spatial heterogeneity — gradients of signalling molecules, localised translation near the membrane, chromosome-excluded cytoplasmic volumes — that well-mixed models cannot see. The paper argues these spatial effects materially influence phenotype.

The cost is substantial: spatial simulation at full resolution requires GPU acceleration and is computationally much more demanding. A full cell cycle in MC4D requires specialist hardware; wcEcoli runs on a laptop in ~10 minutes.

For platelet modelling: Spatial resolution could matter significantly. Platelets have highly organised internal compartments (dense granules, alpha-granules, open canalicular system, mitochondria), and calcium signalling involves spatial waves propagating from stores to the plasma membrane. A spatially resolved approach could model these dynamics more faithfully. However, the added complexity and hardware requirements are a significant practical barrier for initial modelling work.

3.4 Metabolic treatment: FBA vs ODE kinetics

wcEcoli uses flux-balance analysis for metabolism: it finds the optimal flux distribution through the metabolic network at each timestep, given the current enzyme activities and nutrient availability. FBA does not require kinetic rate constants for every reaction, which is an advantage when those constants are unknown. The downside is that FBA assumes steady-state — it cannot model transient metabolic dynamics.

MC4D uses ODE kinetic modelling for metabolism: reactions have explicit rate laws, and metabolite concentrations evolve over time. This captures transient dynamics (e.g., rapid dNTP depletion during replication) but requires kinetic parameters for every reaction.

For platelet modelling: Transient dynamics matter enormously for platelets. Activation happens over seconds to minutes, and the ATP/ADP ratio, calcium concentration, and signalling second messengers all change rapidly. ODE kinetics is the more appropriate formalism for platelet metabolism, and MC4D's approach is better aligned with the biology.

3.5 Chromosome and replication treatment

In wcEcoli, DNA replication is modelled as discrete replisomes advancing along the chromosome at rates determined by nucleotide availability and polymerase speed. Position is tracked as a single coordinate. There is no physical model of the chromosome as a polymer.

MC4D models the chromosome as a Brownian dynamics polymer using LAMMPS molecular dynamics. SMC proteins (condensin analogues) and topoisomerases are represented explicitly and control compaction and segregation. dNTP pools from the metabolic ODE regulate replication speed, creating feedback between metabolism and chromosome dynamics.

For platelet modelling: This difference is largely irrelevant — platelets have no nucleus and no DNA replication. Neither model's chromosome treatment will be used.

3.6 Stochasticity

Both models are stochastic, but MC4D is more thoroughly so. In wcEcoli, some processes (gene regulation, molecule partitioning) are stochastic; others (metabolism via FBA, initiation rules) are deterministic. In MC4D, all molecular processes are governed by stochastic formalisms (RDME or CME), and the result is that every simulated replicate produces a genuinely unique cell — population heterogeneity is an intrinsic output of the model, not a post-hoc analysis.

This matters for platelet biology. Platelets show substantial cell-to-cell heterogeneity in activation thresholds, granule content, and response kinetics. A fully stochastic model would capture this naturally.

3.7 Framework design and reusability

wcEcoli is explicitly architected for reuse. The separation between `wholecell/` (generic engine) and `models/ecoli/` (organism-specific biology) is a deliberate design choice. The framework has already been extended in the literature (colony-level simulations in Skalnik et al. 2023) and is designed so a new organism namespace can be added.

MC4D is a research prototype. The code is open-source and well-documented, but it is not architected as a reusable framework. Using it as a starting point for a different organism would require substantial reworking of the core infrastructure.

3.8 Tooling and accessibility

wcEcoli ships with 80+ analysis scripts, a custom binary data format with reader/writer libraries, and (via this project) a browser-based web UI for running and inspecting simulations without command-line expertise.

MC4D has no equivalent tooling. It produces trajectory files that require custom analysis. There is no web interface, no preset simulation runner, and no built-in analysis scripts. Running it requires GPU hardware, a custom LAMMPS build, and a conda environment with five specialist dependencies.

3.9 What each model is best suited for

Question	Better model
How does growth rate depend on nutrient availability?	wcEcoli
How does gene expression vary across the cell spatially?	MC4D
What is the systems-level effect of a gene knockout?	wcEcoli
How does chromosome segregation affect daughter-cell heterogeneity?	MC4D
How do signalling cascades interact with metabolism?	wcEcoli (closer)
What is the spatial source of cell-to-cell variability?	MC4D
Running many simulations quickly for statistical analysis	wcEcoli
First principles spatial modelling of organelle dynamics	MC4D

3.10 Implications for the platelet model

A platelet whole-cell model sits somewhere between the two approaches in its requirements. Some considerations:

- **Spatial heterogeneity matters.** Calcium waves, granule localisation, and membrane receptor clustering are inherently spatial. MC4D's RDME approach would handle these more faithfully than wcEcoli's well-mixed compartments.
- **Transient dynamics matter.** Platelet activation happens over seconds; FBA's steady-state assumption is unsuitable. ODE kinetics (as in MC4D) is the right formalism for the metabolic and signalling processes.
- **No chromosome needed.** The MC4D chromosome machinery (BD + LAMMPS) is not relevant to a platelet model and would not be carried over.
- **Framework reusability is practical.** Starting from wcEcoli's generic engine is more practical than reworking MC4D for a different organism. The key question is whether spatial resolution can be added later, once a working non-spatial stub exists.

- **A pragmatic path:** build the initial platelet model using the wcEcoli framework (fast, reusable, well-tooled), using ODE kinetics for calcium and signalling rather than FBA. If spatial effects prove important, the RDME approach from MC4D could be adopted for a second-generation model — or the two frameworks could be consulted for methodological guidance even if not directly reused.